The night sky (327-332)

Observing the night sky

1. Approximately how many stars can people see with the naked eye?

2. What does the number of stars you can see depend on?

3. List some differences in the bright points of light in the sky.

4. Other than stars, what could these objects in the sky be?

Stars

1. What are stars?

2. The heat and light from stars comes from nuclear reactions. Describe a nuclear fusion reaction.

3. The nearest star to Earth is the Sun.
   i. How far from Earth is the Sun?

   ii. How long does it take for the heat and light from the Sun to reach Earth?

The Milky Way

1. Why was the Milky Way given its name?

2. What do scientists know about the Milky Way now?

Constellations

1. What is a constellation?

2. What is important about the constellation the Southern Cross?

3. What is the most famous set of constellations in Western culture?

The Zodiac

1. Why are the twelve constellations of the zodiac considered as special?

2. How were the dates for each star sign assigned?

3. What is the astrology?

4. What is astronomy?

Science as a Human Endeavour

People from different cultures identify constellations by relating them to familiar objects. The group of stars in figure 9.1.5 has many interpretations. Describe the interpretation given by the:

- Ancient Greeks

- Modern Australians

- Indigenous Australians in the Northern Territory

Planets

1. How are planets different to stars?

2. List the first four planets.

3. These planets are called the rocky planets. What is another name for them and why are they given this name?

4. When are Mercury and Venus usually visible?

5. What do Venus and Mars look like in the sky?

6. List the four outer planets.

7. What is the structure of these planets?

8. What are they commonly called?

Dwarf Planets

1. When was Pluto discovered?

2. What are dwarf planets and how are they different to ‘normal’ planets?

3. Pluto was reclassified as a dwarf planet. List other dwarf planets that have been found.